

ERZIN, Russian Federation

WMO: 363070

Lat: 50.258N

Lon: 95.171E

Elev: 3611

StdP: 12.88

Time Zone: 7.00 (E07)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -43.3	(c) -39.8	(d) -48.6	(e) 0.4	(f) -42.9	(g) -45.2	(h) 0.5	(i) -39.5	(j) 7.5	(k) -17.2	(l) 6.3	(m) -19.1	(n) 2.2	(o) 110	(p) 0.613

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 21.7	(c) 87.7	(d) 58.9	(e) 84.3	(f) 58.1	(g) 81.2	(h) 57.4	(i) 62.1	(j) 78.7	(k) 60.8	(l) 76.6	(m) 59.6	(n) 75.1	(o) 4.8	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
57.4	80.3	65.2	55.5	75.1	64.9	53.4	69.4	65.3	29.7	78.3	28.7	76.5	27.8	75.7	68.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max	Min		Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.6	11.5	9.8		-45.8	94.2	5.1	2.7	-49.5	96.1	-52.5	97.7	-55.4	99.2	-59.2	101.2
				-46.0	65.2	5.1	2.3	-49.6	66.8	-52.6	68.2	-55.5	69.4	-59.2	71.1
				DB											
				WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.9	-24.8	-17.5	5.7	38.0	50.9	64.3	67.4	63.5	51.5	36.3	10.0	-13.6		
	DBStd	33.89	7.38	10.08	12.68	12.00	8.83	7.10	5.85	6.46	8.54	9.48	10.99	9.36		
	HDD50	9748	2318	1889	1372	380	100	3	0	1	82	430	1201	1972		
	HDD65	13813	2783	2309	1837	810	443	96	42	106	409	890	1651	2437		
	CDD50	1672	0	0	0	20	127	433	538	420	128	6	0	0		
	CDD65	262	0	0	0	0	5	76	116	60	5	0	0	0		
	CDH74	3095	0	0	0	3	127	946	1220	703	96	0	0	0		
	CDH80	917	0	0	0	0	21	289	381	213	13	0	0	0		
	WSAvg	4.3	2.3	2.6	3.2	5.3	6.3	5.8	5.1	4.5	5.0	4.8	3.9	2.8		
	PrecAvg	7.6	0.2	0.2	0.2	0.3	0.5	1.0	1.9	1.6	0.7	0.3	0.4	0.4		
	PrecMax	11.1	0.6	0.5	0.6	0.8	1.0	2.3	4.9	4.7	1.5	1.1	1.0	0.7		
(15) Precipitation	PrecMin	4.7	0.0	0.0	0.0	0.0	0.0	0.2	0.5	0.2	0.1	0.0	0.2	0.0		
	PrecStd	1.9	0.1	0.1	0.2	0.2	0.3	0.4	1.0	1.1	0.4	0.3	0.2	0.2		
	DB	-2.1	15.3	44.7	72.5	84.0	91.2	92.4	91.3	81.9	66.0	42.2	14.9			
	MCWB	-3.4	12.3	34.8	49.1	53.4	61.7	61.5	59.6	55.4	46.0	31.7	13.1			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	-6.7	8.4	36.3	67.1	77.8	86.5	87.9	85.4	76.6	61.0	35.4	8.5			
	2% MCWB	-7.7	6.6	30.8	46.8	52.2	57.7	60.1	58.3	54.0	43.6	28.7	7.2			
	5% DB	-10.2	2.8	29.2	61.3	73.1	83.4	84.1	81.4	71.9	55.6	30.5	3.6			
	5% MCWB	-10.9	1.5	25.9	44.0	50.4	57.0	59.7	58.0	52.0	41.0	26.4	2.7			
	10% DB	-13.5	-2.2	24.0	56.3	68.8	79.6	80.9	77.4	67.1	50.8	26.0	0.0			
	10% MCWB	-14.1	-3.3	21.8	42.0	48.0	55.9	58.4	56.8	50.0	38.6	22.7	-0.7			
	0.4% WB	-3.1	12.7	36.1	50.5	55.8	63.2	65.2	62.9	57.9	46.6	32.6	13.4			
	0.4% MCDB	-2.1	15.4	44.1	70.5	77.8	84.6	80.2	78.4	75.1	64.2	40.7	14.5			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	-7.6	6.7	30.6	47.4	52.9	60.9	62.7	61.4	55.4	43.9	29.4	7.1			
	2% MCDB	-6.8	8.5	36.2	64.8	75.6	78.6	79.9	76.8	74.0	59.4	34.4	8.4			
	5% WB	-10.8	1.6	26.2	44.8	51.0	59.4	61.3	59.9	53.2	41.6	26.6	2.7			
	5% MCDB	-10.2	2.8	29.2	59.7	71.5	75.9	77.0	75.6	69.6	54.0	30.9	3.4			
	10% WB	-14.1	-3.3	21.9	42.3	48.6	58.0	60.2	58.3	51.0	39.1	22.9	-0.6			
	10% MCDB	-13.5	-2.3	24.1	55.0	67.2	74.8	75.1	74.1	65.2	49.7	26.2	0.0			
	MDBR	16.6	20.3	22.3	20.4	23.5	23.3	21.7	22.2	23.2	20.7	16.4	14.7			
	5% DB	17.5	21.6	21.2	28.5	29.7	28.4	27.0	28.5	28.5	26.6	20.4	14.5			
(36) Mean Daily Temperature Range	MCWBR	16.6	20.0	15.7	14.3	13.7	10.1	10.0	10.9	12.6	14.1	14.5	13.7			
	MCDBR	17.4	21.6	20.2	25.7	27.7	22.9	21.7	22.8	26.0	24.9	19.8	14.2			
	MCWBR	16.6	19.8	14.8	13.5	13.6	9.0	9.6	10.1	12.2	13.6	14.0	13.5			
	MCDBR	16.6	19.8	14.8	13.5	13.6	9.0	9.6	10.1	12.2	13.6	14.0	13.5			
(40) Clear-Sky Solar Irradiance	taub	0.242	0.269	0.311	0.351	0.365	0.359	0.362	0.343	0.306	0.290	0.270	0.245			
	taud	2.256	2.209	2.205	2.275	2.268	2.378	2.396	2.421	2.477	2.446	2.315	2.226			
	Ebn at Noon	257	276	282	283	283	285	282	283	284	268	242	234			
	Edh at Noon	22	30	36	38	40	36	36	33	28	23	21	19			
(44) All-Sky Solar Radiation	RadAvg	418	730	1241	1697	2030	2140	1981	1719	1361	855	446	324			
	RadStd	20	34	43	86	86	99	113	95	94	54	33	21			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (5 neighbors)	N/A	N/A	-4.01	N/A	-1.58	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page